

EDA - DATA CLEANING

AIM:

To perform Exploratory Data Analysis (EDA) and data cleaning on the Earthquake dataset to identify patterns, handle missing values, remove duplicates, and prepare the data for Machine Learning.

PROCEDURE:

1. Import the required Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.
2. Load the Earthquake dataset using Pandas.
3. Display the first few records and check the dataset shape.
4. Examine the data types and dataset information.
5. Check for missing or null values.
6. Handle missing values using suitable methods.
7. Identify and remove duplicate records.
8. Check for incorrect or inconsistent data.
9. Perform statistical analysis using describe().
10. Visualize important features using graphs and charts.
11. Analyze correlations between numerical features using a heatmap.
12. Save the cleaned dataset for further analysis and Machine Learning.

CODE:

```
import pandas as pd

# Import Excel file
df = pd.read_excel(r"C:\Users\babyn\Downloads\archive\earthquake_data.xlsx")

# Display first 5 records
print("First 5 Records")
print(df.head())

# Dataset Information
print("\nDataset Information")
df.info()

# Statistical Summary
print("\nStatistical Summary")
print(df.describe(include='all'))
```

OUTPUT:

First 5 Records														
								title	magnitude		date_time	cdi	mmi	alert
tsunami	sig	net	nst	dmin	gap	magType	depth	latitude	longitude					
location	continent			country										
0	M 7.0 - 18 km SW of Malango, Solomon Islands							7.0	22-11-2022 02:03	8	7	green		
1	768 us 117 0.509 17.0 mww 14.000 -9.7963 159.596 Malango, Solomon Islands	Oceania	Solomon Islands											
1	M 6.9 - 204 km SW of Bengkulu, Indonesia							6.9	18-11-2022 13:37	4	4	green		
0	735 us 99 2.229 34.0 mww 25.000 -4.9559 100.738 Bengkulu, Indonesia													
NaN			NaN											
2						M 7.0 -		7.0	12-11-2022 07:09	3	3	green		
1	755 us 147 3.125 18.0 mww 579.000 -20.0508 -178.346													NaN
	Oceania			Fiji										
3	M 7.3 - 205 km ESE of Neiafu, Tonga							7.3	11-11-2022 10:48	5	5	green		
1	833 us 149 1.865 21.0 mww 37.000 -19.2918 -172.129													Neiafu, Tonga
NaN			NaN											
4						M 6.6 -		6.6	09-11-2022 10:14	0	2	green		
1	670 us 131 4.998 27.0 mww 624.464 -25.5948 178.278													NaN
NaN			NaN											

Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 782 entries, 0 to 781
Data columns (total 19 columns):
#   Column      Non-Null Count  Dtype
---  -
0   title       782 non-null    object
1   magnitude   782 non-null    float64
2   date_time   782 non-null    object
3   cdi         782 non-null    int64
4   mmi         782 non-null    int64
5   alert       415 non-null    object
6   tsunami     782 non-null    int64
7   sig         782 non-null    int64
8   net         782 non-null    object
9   nst         782 non-null    int64
10  dmin        782 non-null    float64
11  gap         782 non-null    float64
12  magType     782 non-null    object
13  depth       782 non-null    float64
14  latitude    782 non-null    float64
15  longitude   782 non-null    float64
16  location    777 non-null    object
17  continent   206 non-null    object
18  country     484 non-null    object
dtypes: float64(6), int64(5), object(8)
memory usage: 116.2+ KB
```

Statistical Summary

	magnitude	cdi	mmi	tsunami	sig	nst	dmin
gap	depth	latitude	longitude				
count	782.000000	782.000000	782.000000	782.000000	782.000000	782.000000	782.000000
	782.000000	782.000000	782.000000	782.000000			
mean	6.940153	4.333760	5.964194	0.388747	776.008951	275.396419	2.231937

```

25.807864  75.883168    2.386000   39.734707
std      0.445213    2.658939    1.983592    0.487778   185.083392   218.192074    2.839655
17.652131  137.279603   30.344421   108.016952
min      6.500000    0.000000    1.000000    0.000000   650.000000    0.000000    0.005000
9.000000    9.000000   -61.848000  -179.962000
50%      6.800000    4.000000    6.000000    0.000000   708.500000   213.000000    1.300000
21.000000   26.000000   -3.580000   103.982000
75%      7.100000    7.000000    7.000000    1.000000   819.750000   385.000000    2.880000
29.000000   55.000000   26.126500   139.710250
max      9.100000    9.000000    9.000000    1.000000  2910.000000   934.000000   17.654000
239.000000  670.810000   71.631200   179.662000
[11 rows x 19 columns]

```

```

import pandas as pd

df = pd.read_csv(r"C:\Users\babyn\Downloads\archive\earthquake_data.csv")
print(df.head())

```

OUTPUT:

```

              title magnitude      date_time  cdi  mmi  alert
tsunami sig net nst  dmin  gap magType  depth latitude longitude
location continent      country
0  M 7.0 - 18 km SW of Malango, Solomon Islands      7.0  22-11-2022 02:03    8    7  green
1  768  us  117  0.509  17.0    mww  14.000  -9.7963   159.596  Malango, Solomon Islands
Oceania Solomon Islands
1  M 6.9 - 204 km SW of Bengkulu, Indonesia      6.9  18-11-2022 13:37    4    4  green
0  735  us   99  2.229  34.0    mww  25.000  -4.9559   100.738    Bengkulu, Indonesia
NaN      NaN
2  M 7.0 -      7.0  12-11-2022 07:09    3    3  green
1  755  us  147  3.125  18.0    mww  579.000  -20.0508   -178.346      NaN
Oceania      Fiji
3  M 7.3 - 205 km ESE of Neiafu, Tonga      7.3  11-11-2022 10:48    5    5  green
1  833  us  149  1.865  21.0    mww  37.000  -19.2918   -172.129    Neiafu, Tonga
NaN      NaN
4  M 6.6 -      6.6  09-11-2022 10:14    0    2  green
1  670  us  131  4.998  27.0    mww  624.464  -25.5948   178.278      NaN
NaN      NaN
[5 rows x 19 columns]

```

```

import pandas as pd
import sqlite3

# Read Excel file
df = pd.read_excel(r"C:\Users\babyn\Downloads\archive\earthquake_data.xlsx")

# Connect to SQLite database
conn = sqlite3.connect("earthquake_data.db")

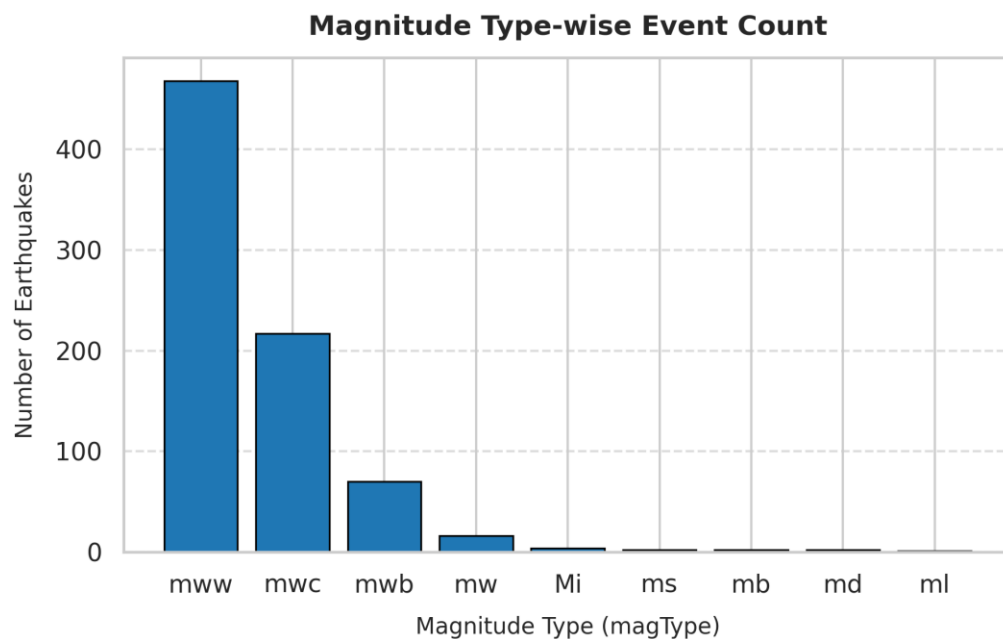
```

```
# Export data to SQLite table
df.to_sql("earthquake_data", conn, if_exists="replace", index=False)
print("Data exported to SQLite successfully!")
conn.close()
```

```
import matplotlib.pyplot as plt

mag_type = df["magType"].value_counts()
plt.bar(mag_type.index, mag_type.values)
plt.title("Magnitude Type-wise Event Count")
plt.xlabel("Magnitude Type (magType)")
plt.ylabel("Number of Earthquakes")
plt.show()
```

OUTPUT:



RESULT:

The Earthquake dataset was successfully cleaned and analyzed using EDA. Missing values and duplicate records were handled, important patterns were identified through visualizations, and the cleaned dataset was prepared for further Machine Learning analysis.